

Week1 Homework

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Prob 1

This algorithm generate all $n!$ permutations with **unequal** probability.

The algorithm will choose $j_1, j_2, \dots, j_n$ independently in $\{1, 2, \dots, n\}$, so there is n^n possibilities. But n^n is not a multiplication of $n!$ when $n \geq 3$ since $n - 1 \mid n!$ but $n - 1 \nmid n^n$.

Prob 2

a. For token embeddings x and y , let $x_1, x_2, \dots, x_d, y_1, y_2, \dots, y_d$ be the coor. of x and y . Then each x_i and y_i are uniformly in $[-1, 1]$. We have

$$\cos(x, y) = \frac{\sum_{i=1}^d x_i y_i}{\sqrt{\sum_{i=1}^d x_i^2 \sum_{i=1}^d y_i^2}}.$$

Obviously $\mathbb{E}(\cos(x, y)) = 0$, since y and $-y$ can be a pair and has contribution 0.

And

$$\text{Var}(\cos(x, y)) = \mathbb{E}(\cos^2(x, y)) = \sum_{i=1}^d \sum_{j=1}^d \mathbb{E} \left(\frac{x_i y_i x_j y_j}{\sum_{k=1}^d x_k^2 \sum_{k=1}^d y_k^2} \right).$$

When $i \neq j$, the expectation is 0 since x_i and $-x_i$ can be a pair. So

$$\begin{aligned} \sum_{i=1}^d \sum_{j=1}^d \mathbb{E} \left(\frac{x_i y_i x_j y_j}{\sum_{k=1}^d x_k^2 \sum_{k=1}^d y_k^2} \right) &= \sum_{i=1}^d \mathbb{E} \left(\frac{x_i^2 y_i^2}{\sum_{k=1}^d x_k^2 \sum_{k=1}^d y_k^2} \right) \\ &= d \cdot \mathbb{E} \left(\frac{x_1^2}{\sum_{k=1}^d x_k^2} \right) \cdot \mathbb{E} \left(\frac{y_1^2}{\sum_{k=1}^d y_k^2} \right) \\ &= \frac{1}{d} \cdot \mathbb{E} \left(\frac{\sum_{k=1}^d x_k^2}{\sum_{k=1}^d x_k^2} \right) \cdot \mathbb{E} \left(\frac{\sum_{k=1}^d y_k^2}{\sum_{k=1}^d y_k^2} \right) \\ &= \frac{1}{d}. \end{aligned}$$

Thus $\mathbb{E}(\cos(x, y)) = 0$, $\text{Var}(\cos(x, y)) = \frac{1}{d}$.

b. In $q_i = Qx_i$, we have $2d^2 - d$ operations, so compute all q_i, k_i, v_i take $6nd^2 - 3nd$ operations.

In $c_{ij} = q_i \cdot k_j$, we have $2d - 1$ operations, so compute all c_{ij} take $2n^2d - n^2$ operations.

In $y_i = \sum_{j=1}^n c_{ij} v_j$, we have $2nd - d$ operations, so compute all y_i take $2n^2d - nd$ operations.

Totally, $4n^2d + 6nd^2 - n^2 - 4nd$ operations.

c. We have $L(4n^2d + 6nd^2 - n^2 - 4nd) = 4e16 + 6e15 - 1e12 - 4e11$ operations for a query. Then total energy is

$$(4e16 + 6e15 - 1e12 - 4e11) * 7e9 * 2e-10 = 5.6e16 + 8.4e15 - 1.4e12 - 5.6e11 \approx 6.44 \times 10^{16} J.$$

Prob 3

a.1. Since only positions that Alice and Bob pick the same basis are retained, obviously their classical bits are the same.

a.2. Since we need their basis are the same, so obtain $\frac{N}{2}$ bits.

b. If Eve picks the right basis, the error rate is 0. If eve picks wrong basis, easy to see both states is with $\frac{1}{2}$ prob. to appear, so the error rate is $\frac{1}{2}$.

Totally, error rate is $\frac{1}{4}$.

c. Prob. of Eve is detected is $\left(\frac{3}{4}\right)^m$, since Eve must be right at every position. As m increases, the prob. will exponentially fast decrease to 0.

Declaration of AI Use

I used AI to translate part of words.